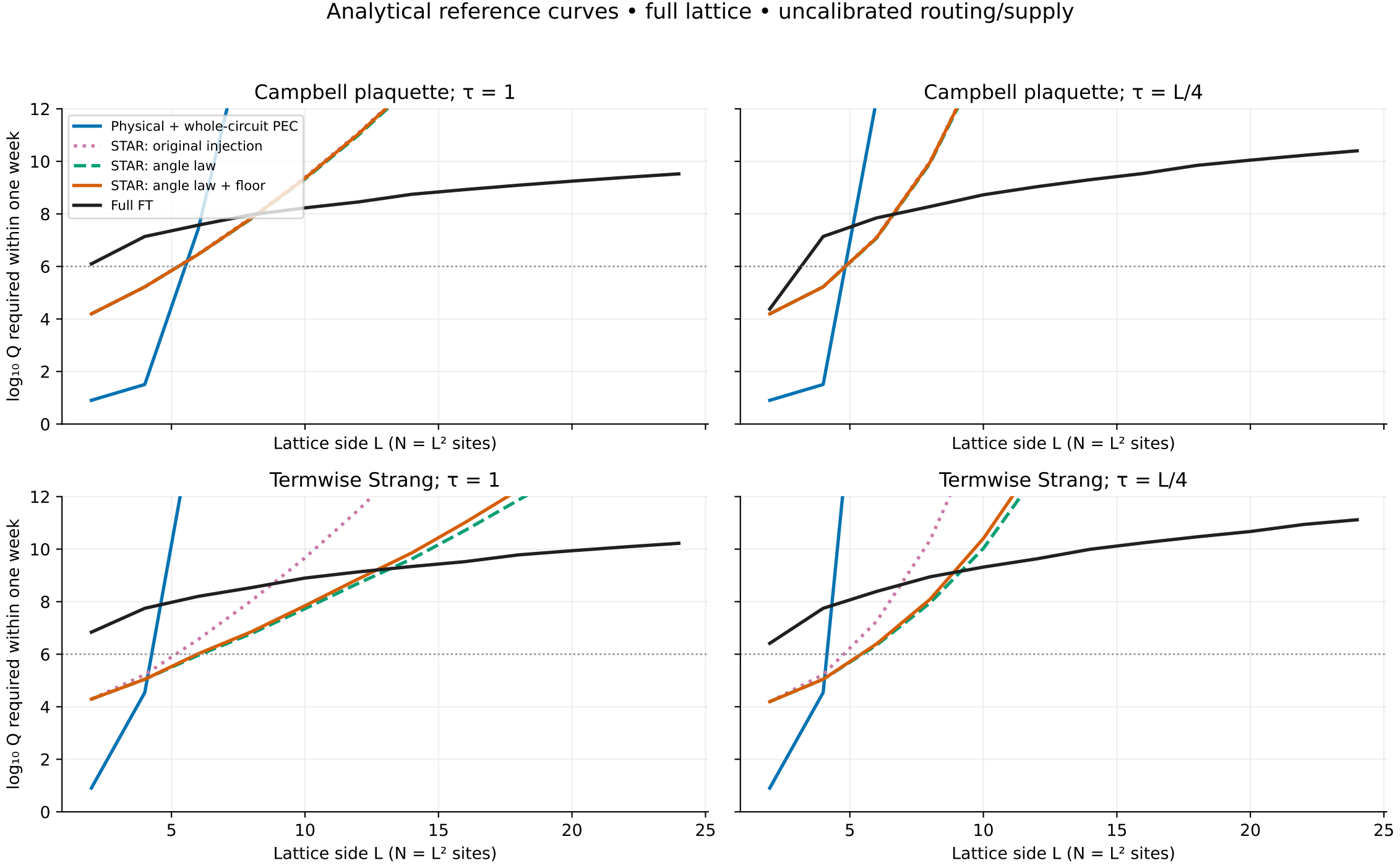
Analytical reference curves • full lattice • uncalibrated routing/supply
Campbell plaquette; τ = 1
Campbell plaquette; τ = L/4
Termwise Strang; τ = 1
Termwise Strang; τ = L/4
Physical + whole-circuit PEC
STAR: original injection
STAR: angle law
STAR: angle law + floor
Full FT
log₁₀ Q required within one week
Lattice side L (N = L² sites)
p = 10⁻³; r = ⌈4τ⌉; 1% relative at D* = 0.1; 95% bounded-sampling criterion
Curves leaving the frame continue above 10¹² qubits; CSV retains their logarithms. Dotted horizontal line: Q = 10⁶.